

Literature Status: Higher-Dimensional Equilateral Sets In Banach–Mazur Compacta

Automatic math-discovery smoke test

July 19, 2026

Status

Already answered in the literature. The higher-dimensional equilateral-set existence question raised in the source paper is answered by a separate later paper.

Source Question

The source paper is Tomasz Kobos and Konrad Swanepoel, *Equilateral dimension of the planar Banach–Mazur compactum*, arXiv:2503.21478 [1]. The relevant passage appears in the Concluding remarks on PDF page 13 and source lines 424–426.

In compact form, the source passage says that it is not clear whether the planar construction can be generalized to higher dimensions. It then states that most likely the n -dimensional Banach–Mazur compactum contains arbitrarily large equilateral sets for every $n \geq 2$.

Supporting Answer

The supporting paper is Florian Grundbacher and Tomasz Kobos, *Exact Banach–Mazur distances of certain ℓ_p -sums and cones*, arXiv:2603.18268 [2]. The authors explicitly connect their result to the source paper: the abstract says that they lift a recent construction of arbitrarily large equilateral sets in the 2-dimensional symmetric compactum to all higher dimensions.

The decisive statement is Corollary 1.9 on PDF page 5:

for every $n, m \geq 2$, there are equilateral sets of cardinality m in $\mathcal{BM}_s^n$.

This directly answers the higher-dimensional existence question in the symmetric Banach–Mazur compactum. The proof of Corollary 1.9 appears near the end of Section 4, on PDF page 23. It applies the local isometric embedding of Corollary 1.8 to the equilateral family from Kobos–Swanepoel.

Scope Limitations

This status packet records an already-known answer to the source’s higher-dimensional existence question. It does not resolve the source paper’s separate planar question about obtaining equilateral sets for every distance in an interval $(1, c]$, nor the broader speculation that the Banach–Mazur compactum might contain all finite metric spaces after scaling distances close to 1.

Search Evidence

The local lightweight indexes

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registry_index.tsv,  solutions/index.tsv,  
attempts/index.tsv,  proof_gaps/index.tsv
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were searched for arXiv:2503.21478, the paper title, “equilateral dimension”, “planar Banach–Mazur”, “Banach–Mazur compactum”, and “equilateral”. The hits were broader equilateral-set packets and attempts, but no duplicate packet for this exact higher-dimensional source question.

A local parsed-source search found arXiv:2603.18268. Its abstract, Corollary 1.9, and proof of Corollary 1.9 explicitly cite/use the planar Kobos–Swanepoel construction and lift it to all dimensions.

Packet Files

The original source PDF is stored as `source_paper.pdf`. The decisive supporting answer PDF is stored as `supporting_paper_2603.18268.pdf`.

References

- [1] T. Kobos and K. Swanepoel, *Equilateral dimension of the planar Banach–Mazur compactum*, arXiv:2503.21478; Proc. Amer. Math. Soc. 153(10) (2025), 4423–4436.
- [2] F. Grundbacher and T. Kobos, *Exact Banach–Mazur distances of certain ℓ_p -sums and cones*, arXiv:2603.18268, 2026.